

STAT 405 Final Project: Proposal

Ratana Wang (45508256), Victor Liu (80536576)

GitHub Repository: <https://github.com/ratanawang/stat405>

Car Crashes and Pedestrians in Vancouver

Datasets + Data Cleaning

The two main datasets we used are:

- [this Kaggle dataset, containing 2018-2022 data](#) on car crashes reported to ICBC, combined with the Tableau dataset containing [official ICBC 2020-2024 data](#). We removed overlapping data (i.e., 2020-2022) from one of the datasets before combining them into one larger dataset. With the 2018-2024 data, we can gather a more complete picture of both pre- and post-COVID-19 pandemic trends. Instructions to access the CSV files are below:
 - For the Kaggle dataset, click Download (top right) -> Download dataset as zip.
 - For the Tableau dataset, go to the “ICBC reported crashes” tab (4th tab), then click “All Crash Data”. Then click Show Fields (top right) and select All. Then click the Download icon (also top right corner) -> click Data. In the newly opened window, select Full Data on the left. Then click Download on the top right, which should then download a file called “British Columbia_Full Data_data.csv”
- Government of Canada weather data from 2018-2024 (each year below has a different link; from each link, there is an option to download as CSV). The weather data was taken at Vancouver International Airport. We combined the weather from all these years into one dataframe in our EDA.
 - [2018](#), [2019](#), [2020](#), [2021](#), [2022](#), [2023](#), [2024](#)

We took the combined car crash and weather datasets, removed some columns we don't need, and filtered out rows that contain NA (or equivalent) values for latitude/longitude. Also, we are focusing on the city of Vancouver so that our weather trends are more accurate. We use this clean data for EDA below.

Backup Dataset

Our backup dataset will be car crash data from a different region; specifically, here is a [Kaggle dataset on car crashes in Chicago from 2013 to 2023](#). If we need to use this, we will look for a Chicago weather dataset to pair with it. We might also need to choose a different research question, because there is no “pedestrian involvement” indicator that we can use to tell if a pedestrian was involved in the crash.

Exploratory Data Analysis

Here are screenshots of the `head` outputs of our clean data:

ICBC car crash data (city of Vancouver only):

```
head(crash_cov)
```

```
##   Date.Of.Loss.Year Crash.Severity Cyclist.Flag Day.Of.Week
## 1      2018 CASUALTY CRASH           No    FRIDAY
## 2      2018 CASUALTY CRASH           No    FRIDAY
## 3      2018 CASUALTY CRASH           No    FRIDAY
## 4      2018 CASUALTY CRASH           No    MONDAY
## 5      2018 CASUALTY CRASH           No    MONDAY
## 6      2018 CASUALTY CRASH           No    MONDAY
##   Derived.Crash.Configuration Heavy.Veh.Flag Intersection.Crash Motorcycle.Flag
## 1              REAR END                No                No                No
## 2          UNDETERMINED                No                No                No
## 3              REAR END                No                No                No
## 4 SIDE SWIPE - SAME DIRECTION                No                No                No
## 5              REAR END                No                No                No
## 6              SIDE IMPACT                No                No                No
##   Parked.Vehicle.Flag Parking.Lot.Flag Pedestrian.Flag
## 1              Yes                No                No
## 2              Yes                No                No
## 3              Yes                No                No
## 4              Yes                No                No
## 5              Yes                No                No
## 6              Yes                No                No
##   Street.Full.Name..ifnull. Time.Category Municipality.Name
## 1              W 23RD AVE    15:00-17:59      VANCOUVER
## 2              CAMBIE ST    15:00-17:59      VANCOUVER
## 3      W HASTINGS ST    21:00-23:59      VANCOUVER
## 4      PRINCESS AVE    15:00-17:59      VANCOUVER
## 5      ORMIDALE ST    18:00-20:59      VANCOUVER
## 6      W 8TH AVE    18:00-20:59      VANCOUVER
##   Road.Location.Description Street.Full.Name Total.Crashes Total.Victims
## 1              W 23RD AVE      W 23RD AVE          1          1
## 2              CAMBIE ST      CAMBIE ST          1          1
## 3      W HASTINGS ST      W HASTINGS ST          1          1
## 4      PRINCESS AVE      PRINCESS AVE          1          1
## 5      ORMIDALE ST      ORMIDALE ST          1          1
## 6      W 8TH AVE      W 8TH AVE          1          1
##   Cross.Street.Full.Name Latitude Longitude Mid.Block.Crash yearmonth
## 1              49.25004 -123.1114                N 2018.833
## 2              49.22277 -123.1165                N 2018.833
## 3              49.28756 -123.1183                N 2018.583
## 4              49.27821 -123.0916                N 2018.333
## 5              49.23457 -123.0248                N 2018.417
## 6              49.26378 -123.1058                N 2018.083
```

Weather data (from Vancouver International Airport):

```
head(weather)
```

```
##   Date.Time Year Month Day Max.Temp...C Min.Temp...C Mean.Temp...C
## 1 2018-01-01 2018     1   1         3.0        -4.5         -0.8
## 2 2018-01-02 2018     1   2         3.0        -3.3         -0.2
## 3 2018-01-03 2018     1   3         4.0        -1.6          1.2
## 4 2018-01-04 2018     1   4         5.7         1.4          3.6
## 5 2018-01-05 2018     1   5         9.7         4.6          7.2
## 6 2018-01-06 2018     1   6         9.2         5.0          7.1
##   Total.Rain..mm. Total.Snow..cm. Total.Precip..mm. Snow.on.Grnd..cm.
## 1              0.0              0              0.0              NA
## 2              0.0              0              0.0              NA
## 3              0.0              0              0.0              NA
## 4              0.0              0              0.0              NA
## 5             21.8              0             21.8              NA
## 6              1.0              0              1.0              NA
##   Dir.of.Max.Gust..10s.deg. Spd.of.Max.Gust..km.h. yearmonth
## 1                      NA                      NA      2018
## 2                      NA                      NA      2018
## 3                      NA                      NA      2018
## 4                      NA                      NA      2018
## 5                      NA                      NA      2018
## 6                      29                      43      2018
```

Further EDA can be found here: [Link to EDA \(on GitHub\)](#)

Here is a screenshot of the **head** of our backup data (Chicago car crashes):

```
head(crash_chicago)
```

```
##          CRASH_DATE POSTED_SPEED_LIMIT TRAFFIC_CONTROL_DEVICE
## 1 07/12/2023 03:05:00 PM                30          NO CONTROLS
## 2 07/12/2023 05:50:00 PM                30          NO CONTROLS
## 3 07/12/2023 02:00:00 PM                30          NO CONTROLS
## 4 07/12/2023 07:05:00 AM                30      TRAFFIC SIGNAL
## 5 07/12/2023 06:30:00 PM                30          NO CONTROLS
## 6 08/05/2022 09:25:00 AM                30          NO CONTROLS
##  DEVICE_CONDITION    WEATHER_CONDITION LIGHTING_CONDITION FIRST_CRASH_TYPE
## 1      NO CONTROLS      CLEAR          DAYLIGHT      TURNING
## 2      NO CONTROLS      CLEAR          DAYLIGHT      REAR END
## 3      NO CONTROLS      CLEAR          DAYLIGHT  OTHER OBJECT
## 4      UNKNOWN FREEZING RAIN/DRIZZLE    DAYLIGHT      REAR END
## 5      NO CONTROLS      CLEAR          DAYLIGHT      REAR END
## 6      NO CONTROLS      CLEAR          DAYLIGHT      TURNING
## TRAFFICWAY_TYPE LANE_CNT      ALIGNMENT ROADWAY_SURFACE_COND ROAD_DEFECT
## 1      NOT DIVIDED      NA STRAIGHT AND LEVEL      UNKNOWN NO DEFECTS
## 2      NOT DIVIDED      NA STRAIGHT AND LEVEL      DRY NO DEFECTS
## 3      NOT DIVIDED      NA STRAIGHT AND LEVEL      DRY RUT, HOLES
## 4      NOT DIVIDED      NA STRAIGHT AND LEVEL      UNKNOWN UNKNOWN
## 5      NOT DIVIDED      NA STRAIGHT AND LEVEL      WET NO DEFECTS
## 6 T-INTERSECTION      NA STRAIGHT AND LEVEL      DRY NO DEFECTS
##          REPORT_TYPE      CRASH_TYPE INTERSECTION_RELATED_I
## 1 NOT ON SCENE (DESK REPORT) NO INJURY / DRIVE AWAY
## 2 NOT ON SCENE (DESK REPORT) NO INJURY / DRIVE AWAY
## 3 NOT ON SCENE (DESK REPORT) NO INJURY / DRIVE AWAY
## 4 NOT ON SCENE (DESK REPORT) NO INJURY / DRIVE AWAY
## 5 NOT ON SCENE (DESK REPORT) NO INJURY / DRIVE AWAY
## 6 NOT ON SCENE (DESK REPORT) NO INJURY / DRIVE AWAY Y
## NOT_RIGHT_OF_WAY_I HIT_AND_RUN_I      DAMAGE      DATE_POLICE_NOTIFIED
## 1                                OVER $1,500 07/15/2023 11:30:00 AM
## 2                                $501 - $1,500 07/12/2023 06:41:00 PM
## 3                                Y                                $501 - $1,500 07/21/2023 10:10:00 AM
## 4                                $501 - $1,500 07/12/2023 02:18:00 PM
## 5                                OVER $1,500 07/12/2023 07:15:00 PM
## 6                                OVER $1,500 08/05/2022 10:20:00 AM
## PRIM_CONTRIBUTORY_CAUSE      SEC_CONTRIBUTORY_CAUSE STREET_NO
## 1 IMPROPER TURNING/NO SIGNAL      UNABLE TO DETERMINE      4754
## 2 FOLLOWING TOO CLOSELY      NOT APPLICABLE      8300
## 3 NOT APPLICABLE      NOT APPLICABLE      9615
## 4 FOLLOWING TOO CLOSELY      UNABLE TO DETERMINE      2370
## 5 FOLLOWING TOO CLOSELY      FOLLOWING TOO CLOSELY      5200
## 6 IMPROPER TURNING/NO SIGNAL FAILING TO REDUCE SPEED TO AVOID CRASH      1650
## STREET_DIRECTION      STREET_NAME BEAT_OF_OCCURRENCE PHOTOS_TAKEN_I
## 1      W      63RD ST      813
```

```

## 2          S          PULASKI RD          834
## 3          S STONY ISLAND AVE          431
## 4          N          ASHLAND AVE          1811
## 5          N          ELSTON AVE          1623
## 6          N          THROOP ST          1433
## STATEMENTS_TAKEN_I DOORING_I WORK_ZONE_I WORK_ZONE_TYPE WORKERS_PRESENT_I
## 1
## 2
## 3
## 4
## 5
## 6
## NUM_UNITS      MOST_SEVERE_INJURY INJURIES_TOTAL INJURIES_FATAL
## 1          2 NO INDICATION OF INJURY          0          0
## 2          2 NO INDICATION OF INJURY          0          0
## 3          1 NO INDICATION OF INJURY          0          0
## 4          2 NO INDICATION OF INJURY          0          0
## 5          2 NO INDICATION OF INJURY          0          0
## 6          2 NO INDICATION OF INJURY          0          0
## INJURIES_INCAPACITATING INJURIES_NON_INCAPACITATING
## 1          0          0
## 2          0          0
## 3          0          0
## 4          0          0
## 5          0          0
## 6          0          0
## INJURIES_REPORTED_NOT_EVIDENT INJURIES_NO_INDICATION INJURIES_UNKNOWN
## 1          0          2          0
## 2          0          2          0
## 3          0          1          0
## 4          0          2          0
## 5          0          2          0
## 6          0          2          0
## CRASH_HOUR CRASH_DAY_OF_WEEK CRASH_MONTH LATITUDE LONGITUDE
## 1          15          4          7 41.77854 -87.74206
## 2          17          4          7 41.74213 -87.72182
## 3          14          4          7 41.71984 -87.58479
## 4          7          4          7 41.92510 -87.66829
## 5          18          4          7 41.97526 -87.75199
## 6          9          6          8 41.91222 -87.66191

```

Our Research Question

The main scientific question we want to approach using Bayesian methods is: **given a car crash at a particular intersection, the time of day, and the weather conditions, what is the probability that a pedestrian is hit?**

We looked into existing car crash location analyses and found that they reported simple crash counts based on location (e.g. [this map](#) and [this map](#)), but did not take into account other conditions (weather, time, etc.). Our research question attempts to fill this gap and, by

focusing on pedestrian safety, we hope to surface results that will help Vancouver become a more walkable city.

Bayesian Methods / Research Methodology

Our main parameter of interest is the conditional probability that a pedestrian is involved in a car crash, given other contextual variables, e.g. the intersection (latitude/longitude), weather conditions, and time of day. This parameter will have some prior distribution, and after incorporating the data its posterior distribution will be used to predict pedestrian safety under specific conditions.

Since our data will be quite high-dimensional, we don't expect to be able to use exact inference methods, and instead we'll likely rely on approximate inference methods such as MCMC. We will use R and Stan to conduct our investigations.

We took a brief look at the topics following Quiz 2 and—although we don't really understand these topics yet—we think that potentially “kernel mixtures” could be applicable to our project, since we expect our posterior to look like many different Gaussians centered at each road intersection. For predictive purposes, we could also maybe use “stochastic gradient descent” to find posterior minima (i.e. points in the data space where pedestrians are least likely to be hit by cars).

Notes to the teaching team:

- *If possible, we'd like some help with figuring out which topics (before and after Quiz 2) are most relevant for our project and how to apply them. Thank you!*

Contribution Plan

We plan to work on this project collaboratively, and discuss all plans before proceeding. At the time of writing this project proposal, our team has roughly followed the process: meet -> divide work -> work -> review -> repeat, and it seems to be working for us. We will resolve any conflicts by discussing them, and we will ensure roughly equal contribution by agreeing to communicate well if we feel we are working too much/little.